



BRENT OZAR
UNLIMITED®

Faster, Cheaper Cloud Databases

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Get the slides & scripts

14days.brentozar.com/faster.zip

Download it within 14 days – it'll disappear

Access problems: email Help@BrentOzar.com,
and I will respond to those after the class (not during)

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There's a present in that zip!

You get these classes for a year for free:

- Mastering Server Tuning
- Running SQL Server in Azure & AWS

Follow the instructions in the readme.txt file.

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3 performance metrics

1. How much data are we hosting?
**Total data size in GB/TB,
and number of databases.**
2. How fast are we going?
Batch Requests per second.
3. How hard are we working to achieve that speed?
**Wait time ratio over time
(like days of waits in a 1-day time span)**

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3 ways to track those metrics

1. Third party monitoring software
(Solarwinds, Quest, Idera, Redgate, etc.)
2. Build it yourself with your cloud provider's tools
(AWS Cloudwatch, Azure Managed Grafana, etc.)
3. sp_BlitzFirst fromFirstResponderKit.org
(check out my free How I Use the First Responder Kit class for details on logging it to tables)
- ~~4. Cloud vendor portal defaults~~

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3 categories for those metrics

	Small	Medium	Large
1. Total size, GB	0-250	250-1TB	>1TB
2. Batch req/sec	0-10	10-100	>100
3. Wait time ratio	0-0.25	0.25-1	>1

But your own databases might be a combination of different colors for different metrics.

You just wanna know roughly what group you're in.

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We have 3 dials to handle problems.



Table & Index Design



Query Design



Hardware & Licensing

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PAGEIOLATCH your top wait?

1. Tune indexes, looking at missing indexes:
`sp_BlitzIndex @GetAllDatabases = 1`
2. Tune queries:
`sp_BlitzCache @SortOrder = 'reads'`
3. Add more memory
4. Make storage faster

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Making cloud storage faster is hard

From 2000s to 2020s,
RAM & storage improved 20X-1000X

Cloud hardware regressed back to the 2000s

To handle it, we have to use 2000s-era techniques:

- Get pools of drives for speed, not 1 for capacity
- Pick your VM instance type based on throughput

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SOS_SCHEDULER_YIELD on top?

`sp_BlitzCache @SortOrder = 'cpu'`

Focus on these query issues that burn CPU time:

- String processing (LIKE '%STRING%')
- Row-based processing (cursors, functions)
- Sorting/grouping (but index for that, not reads)
- Heaps with forwarded fetches (rebuild, add CX)
- Implicit conversions

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Picking the right instance types

New cheaper VM types come out all the time with faster CPUs, more memory, more storage, etc:

<https://instances.vantage.sh/> - AWS VMs

<https://instances.vantage.sh/rds/> - AWS RDS

<https://instances.vantage.sh/azure/> - VMs

Azure SQL DB, Managed Instances, Hyperscale:
you're stuck with the ol' documentation

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You can do this.

For questions, leave comments on the relevant module.

For private help after the class, email Help@BrentOzar.com with:

- A note that you were in this class
- sp_Blitz @CheckServerInfo = 1
- sp_BlitzFirst @SinceStartup= 1

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Thanks, and I hope you
had a great time!

We'll take a 5-minute break
so folks can pack up and leave,
then switch to open Q&A.

(Remember, no private questions:
you can't line up to ambush me
after the open Q&A finishes.)

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